ES 404: Networks and Complex Systems

India's District Adjacency Network and COVID-19 Propagation: A Centrality Analysis

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Abstract

Geographic adjacency networks encode the pathways along which infectious diseases propagate across administrative boundaries. This study constructs and analyses a district-level adjacency network of India comprising **666 nodes** (districts and union territories) and **1,784 edges** derived from polygon intersection of the GADM Level-2 boundary shapefile. Four centrality measures (degree, betweenness, eigenvector, and closeness) are computed and correlated with the number of days between India's first reported COVID-19 case (30 January 2020) and each district's first confirmed case, using covid19india.org data spanning April 2020 to June 2021. Louvain community detection identifies **13 mesoscale communities** that are geographically coherent but cross state boundaries, suggesting that epidemic spread followed natural geographic corridors rather than administrative lines. All four centrality measures show *negative* correlations with time-to-first-case (closeness: $r = -0.28$; degree: $r = -0.20$; betweenness: $r = -0.11$; eigenvector: $r = -0.06$), confirming that structurally central districts were reached sooner. The Siliguri Corridor emerges as the dominant chokepoint. This narrow strip in West Bengal, made up of Darjeeling, Jalpaiguri, Koch Bihar, and Uttar Dinajpur, carries betweenness centrality values 40 to 65% higher than any other region, owing to its unique role as the sole land bridge connecting North-East India to the rest of the country. An interactive multi-scenario dashboard extends the network analysis to five policy-relevant scenarios: epidemic spread, railway network resilience, healthcare accessibility, disaster relief logistics, and rural connectivity.

Keywords: geographic adjacency network, district-level epidemiology, COVID-19 propagation, network centrality, Louvain community detection, Siliguri Corridor, India.

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1 Executive Summary

This project applies network science to a geographically defined problem: *how does India's district adjacency structure govern the speed and pattern of COVID-19 propagation?*

Using open-source boundary data and public COVID-19 case records, we:

1. Construct a 666-node district adjacency network from GADM Level-2 shapefiles, including all districts of Jammu and Kashmir and the Ladakh region (Leh and Kargil).

2. Characterise the network topology as a *near-planar lattice* with unimodal degree distribution (mean ≈ 5.36), confirming that India's district network is *not* scale-free.

3. Demonstrate that closeness centrality is the strongest predictor of time-to-first-COVID-case ($r = -0.28$, $n = 517$), while eigenvector centrality is the weakest ($r = -0.06$).

4. Identify the Siliguri Corridor as the single most critical structural chokepoint (top-4 betweenness nodes all in West Bengal's North district belt).

5. Detect 13 Louvain communities whose boundaries are driven by geographic corridors, river systems, and mountain ranges rather than state administrative lines.

2 Introduction

2.1 Motivation

Infectious disease surveillance in India has historically relied on administrative boundaries that do not reflect actual population flow [1]. Districts share personnel, markets, and transport links with their geographic neighbours irrespective of state borders. A *network-theoretic* approach, which treats each district as a node and each shared border as an edge, makes explicit the latent infrastructure through which pathogens travel.

The COVID-19 pandemic provides a uniquely complete natural experiment: nearly every district reported its first case within a 14-month window (April 2020 to June 2021), allowing direct comparison of structural network position against empirical infection timing.

2.2 Research Questions

- **RQ1:** What is the topology of India's district adjacency network, and does it exhibit scale-free properties?
- **RQ2:** Which districts are structurally critical (high betweenness), and do they form identifiable geographic chokepoints?
- **RQ3:** How strongly does a district's network centrality predict the speed at which COVID-19 reached it?
- **RQ4:** Do Louvain communities correspond to administrative state boundaries, or to geographic corridors?

2.3 Scope and Limitations

The analysis uses *geographic adjacency* as a proxy for inter-district contact intensity. This captures topological structure but not population density, transport volume, or cross-border mobility. Betweenness centrality is approximated using $k = 200$ samples for computational tractability. Several J&K districts (Badgam, Bandipore, Kargil, Leh, Poonch, Shopian) have no COVID data in the source dataset and are retained as structurally present nodes with null case counts. Post-2019 administrative reorganisation (Ladakh as a separate Union Territory) is noted, but boundary data follows pre-reorganisation GADM shapes that include both Leh and Kargil.

3 Data and Methods

3.1 Data Sources

Table 1: Data sources used in this study.

Dataset	Source	Description
District boundaries	GADM v4.1 Level-2 (gadm.org)	676 polygon features (666 distinct districts; 10 are multi-polygon), including J&K and Ladakh
COVID-19 cases	covid19india.org	District-level confirmed cases, April 2020 to June 2021

3.2 Network Construction

Districts were treated as nodes. Two districts were connected by an undirected edge if their polygon boundaries intersected or touched (with a 500-metre buffer to capture districts sharing only a point or very short border segment). Intersection tests used the Shapely STRtree spatial index in the UTM Zone 43N projection (EPSG:32643) for metric accuracy.

3.3 Centrality Measures

Four standard centrality measures were computed:

- **Degree centrality**: normalised fraction of neighbours, $C_D(v) = \deg(v)/(N-1)$.
- **Betweenness centrality**: fraction of shortest paths passing through v, approximated with $k = 200$ random node samples [2].
- **Eigenvector centrality**: the leading eigenvector of the adjacency matrix, capturing influence via well-connected neighbours. Eigenvector centrality is computed on the full graph; because the network has 7 connected components, eigenvector mass concentrates on the largest component and is near-zero elsewhere. This is visible as the pileup at the origin of the eigenvector panel in Figure 3 and partly explains the very weak observed correlation.
- **Closeness centrality**: reciprocal of average shortest-path length to all other nodes in the largest connected component ($N_{\text{LCC}} = 657$ nodes).

3.4 Community Detection

Louvain modularity maximisation [3] was applied to the full graph with `random_state=42` for reproducibility. The algorithm identifies communities by iteratively reassigning nodes to maximise:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

where A_{ij} is the adjacency matrix, k_i is node degree, m is edge count, and $\delta(c_i, c_j)$ is 1 if nodes i, j share a community.

3.5 Correlation Analysis

Days from India's index case (30 January 2020) to each district's first confirmed COVID-19 case was used as the outcome variable. Pearson correlation coefficients were computed

between each centrality measure and this timing variable for the 517 districts with complete data. Of the 522 districts with case data, 5 lacked a verifiable first-case date and are excluded from the correlation analysis, leaving $n = 517$.

4 Results

4.1 Network Topology

Table 2: Summary statistics for the India district adjacency network.

Property	Value
Nodes (districts / UTs)	666
Edges (shared borders)	1,784
Average degree	5.36
Maximum degree	11
Minimum degree	0
Degree standard deviation	1.77
Connected components	7
Largest connected component	657 nodes
Average clustering coefficient	0.463
Louvain communities	13
Districts with COVID data	522 / 666
J&K districts in network	22

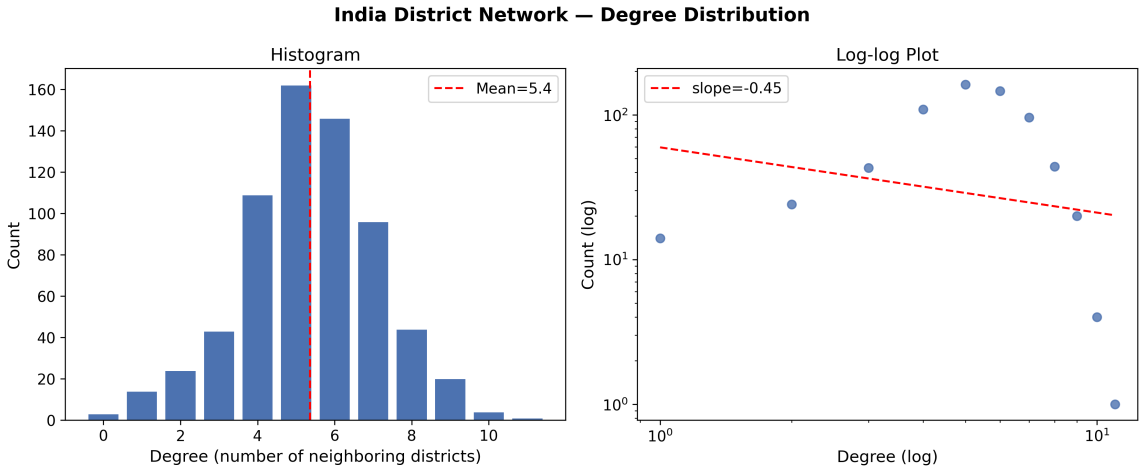


Figure 1: Degree distribution of India's 666-district adjacency network. Left: histogram showing a unimodal peak around degrees 5 and 6, consistent with a near-planar geographic lattice. Right: log-log plot showing no evidence of a power-law tail, ruling out scale-free topology. Mean degree = 5.36 (red dashed line).

The degree distribution (Figure 1) is sharply unimodal and approximately Gaussian, with no heavy tail. This is the expected topology for a geographically embedded lattice network: each district has a small number of physical neighbours determined by its polygon boundary. The network is *not scale-free*; the log-log plot shows a strongly negative and non-linear pattern rather than a straight power-law line.

Seven connected components exist because island territories (Andaman & Nicobar, Lakshadweep) and some union territories share no land border with any other district.

4.2 Geographic Network Structure and Critical Nodes

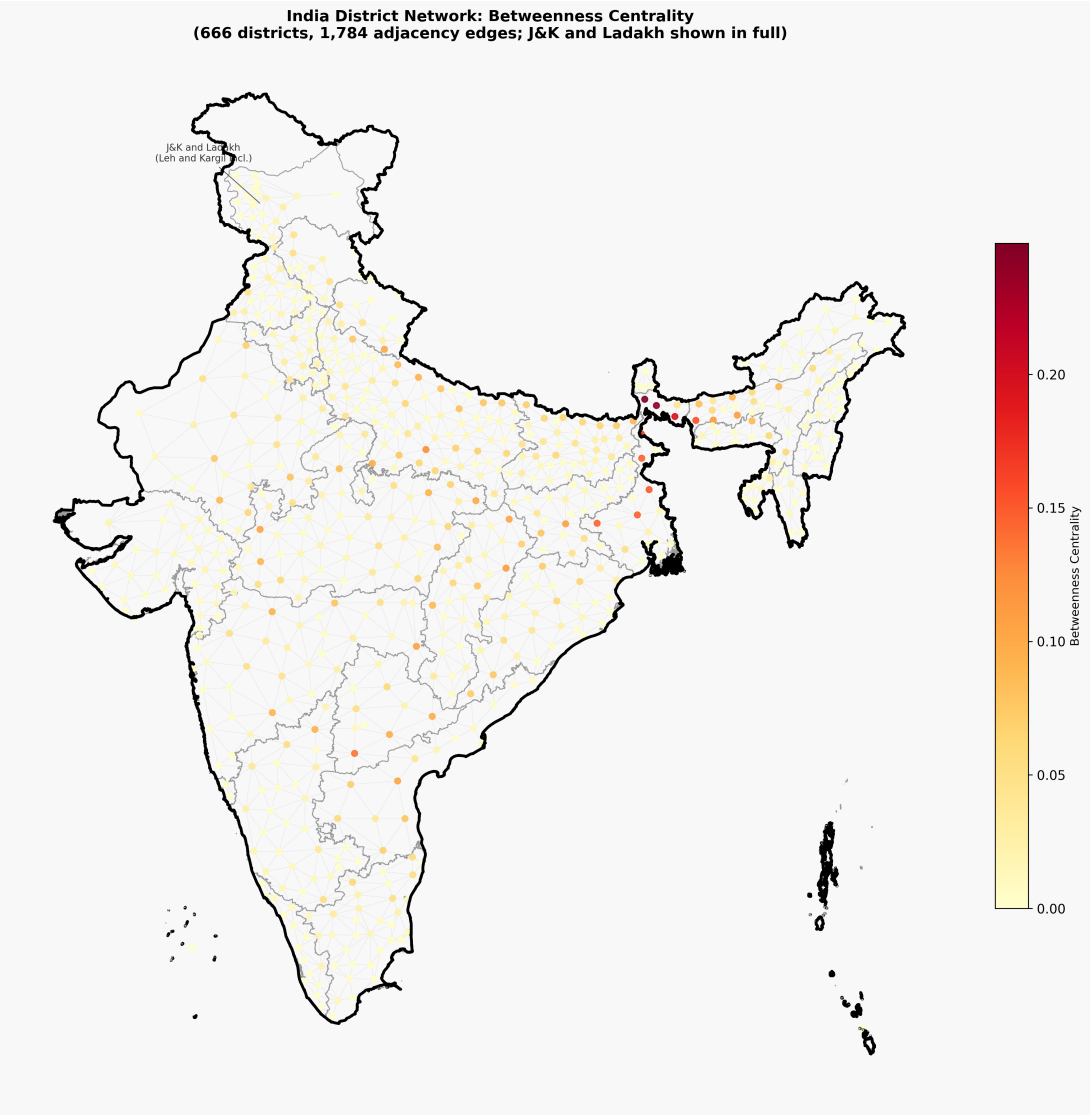


Figure 2: Geographic layout of India's district adjacency network, with nodes coloured by betweenness centrality (yellow = low, red = high). Nodes are placed at district centroids; edges represent shared borders. State/UT outlines are shown. The Siliguri Corridor districts in northern West Bengal are the dominant high-betweenness cluster.

Figure 2 shows the network overlaid on India's geography. High-betweenness districts concentrate in two regions:

1. **The Siliguri Corridor (North West Bengal):** Darjeeling, Jalpaiguri, Koch Bihar, and Uttar Dinajpur dominate the top-4 betweenness ranking. These districts form the only land connection between the Seven Sister states of North-East India and the rest of the country. Any path from North-East India to peninsular India must pass through this corridor.

2. **Central India transition zone:** Districts spanning the Chhattisgarh, Odisha, and Telangana tri-junction show elevated betweenness as traffic between peninsular and central India funnels through this region.

Table 3: Top 10 districts by betweenness centrality. The Siliguri Corridor dominates. All values normalised to $[0, 1]$.

District	**State**	**Betweenness**
Darjeeling	West Bengal	0.2491
Jalpaiguri	West Bengal	0.2392
Koch Bihar	West Bengal	0.1956
Uttar Dinajpur	West Bengal	0.1672
Dhubri	Assam	0.1536
Murshidabad	West Bengal	0.1492
Barddhaman	West Bengal	0.1452
Maldah	West Bengal	0.1437
Puruliya	West Bengal	0.1424
Mahbubnagar	Telangana	0.1356

4.3 Centrality as a Predictor of COVID-19 Arrival Time

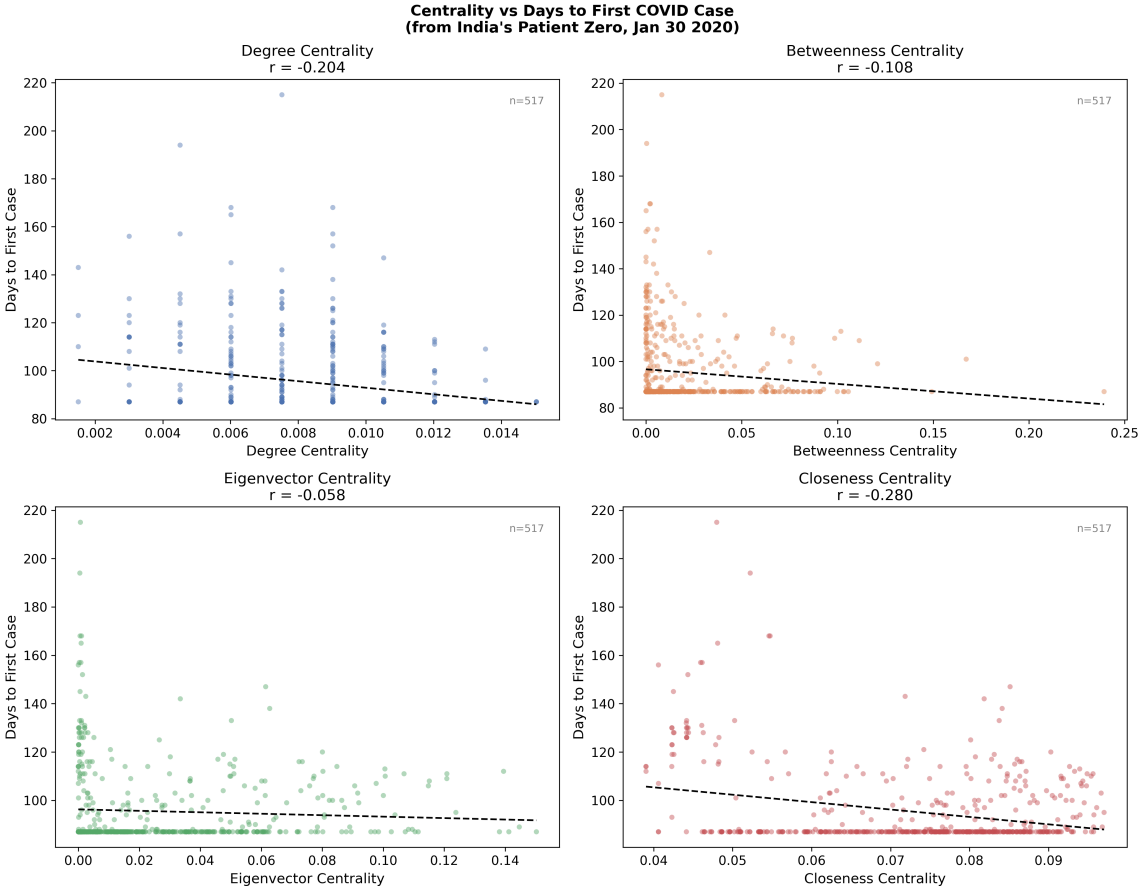


Figure 3: Scatter plots of four centrality measures against days-to-first-COVID-case (from India's index case on 30 January 2020). Each point is one district ($n = 517$). Black dashed lines are OLS regression fits. All correlations are negative: districts with higher centrality received the virus sooner.

All four centrality measures correlate negatively with time-to-first-case (Figure 3, Table 4), confirming that structurally central districts were infected earlier.

Table 4: Pearson correlations between centrality and days-to-first-COVID-case ($n = 517$ districts with complete data).

Centrality Measure	r	**Interpretation**
Closeness centrality	-0.280	Strongest predictor
Degree centrality	-0.204	Moderate predictor
Betweenness centrality	-0.108	Weak predictor
Eigenvector centrality	-0.058	Very weak predictor

Closeness centrality is the strongest predictor ($r = -0.28$). This is interpretable: a district that is *geometrically close* to all other districts in graph distance will

be reached early simply because the virus does not have far to travel from any direction. Degree centrality ($r = -0.20$) also performs well; having more neighbours means more independent routes for importation. Betweenness ($r = -0.11$) and eigenvector ($r = -0.06$) centrality are weaker predictors, suggesting that being a *transit hub* or having *influential neighbours* matters less for early infection than simply being well-connected and geographically central.

The relatively modest $|r|$ values (all below 0.30) reflect the reality that COVID propagation was also driven by airline networks, pilgrimage events, and population density, factors that are not captured by geographic adjacency alone.

4.4 Community Structure

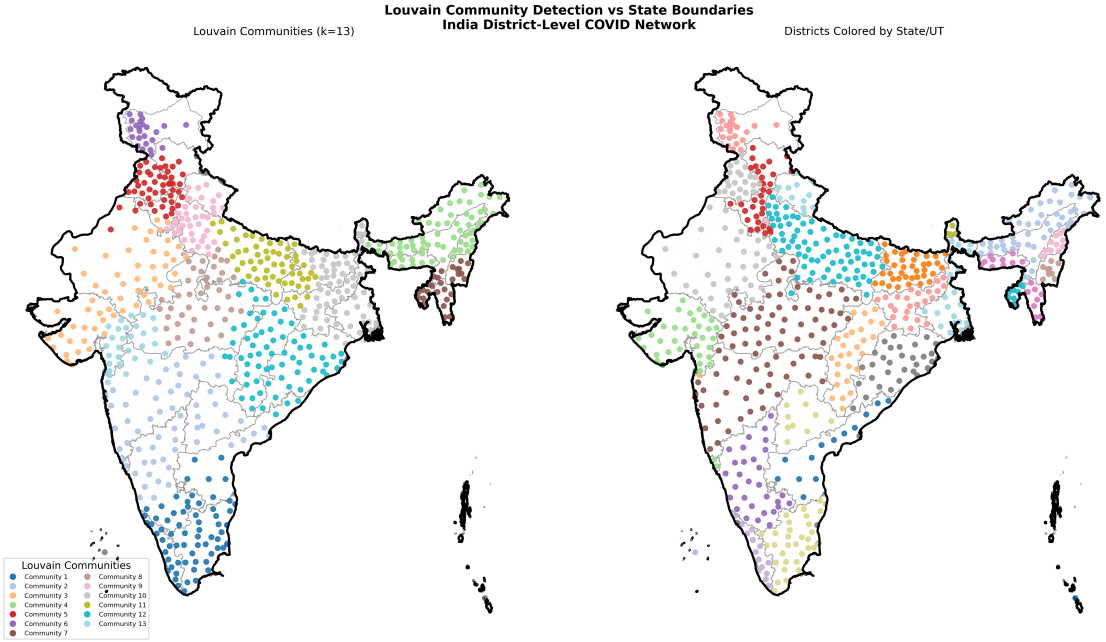


Figure 4: Left: 13 Louvain communities detected in the India district adjacency network ($k = 13$, modularity maximisation with random seed 42). Right: districts coloured by their administrative state/UT for comparison. Community boundaries (left) generally follow geographic corridors and terrain features, often crossing administrative state lines.

Louvain community detection finds **13 communities** whose boundaries are driven by geographic features (Figure 4):

- **North-East cluster**: The Seven Sister states form a natural geographic community nearly isolated from the rest of India. The Siliguri Corridor nodes are the only high-betweenness bridge.
- **Peninsular communities**: The Deccan plateau separates distinct communities in Kerala/Tamil Nadu, coastal Andhra Pradesh, and interior Karnataka/Maharashtra.

- **Gangetic plain**: Dense, high-degree districts in Uttar Pradesh, Bihar, and West Bengal form a large central community due to their continuous border-sharing.
- **Desert/arid community**: Rajasthan and Gujarat districts cluster together, connected via low-population-density adjacency.
- **Himalayan fringe**: Himachal Pradesh, Uttarakhand, and adjacent hill districts form a thin northern arc community.

Critically, community boundaries frequently *cross state lines*, demonstrating that the effective epidemiological neighbourhoods for COVID spread were geographic, not administrative.

4.5 COVID-19 Case Choropleth

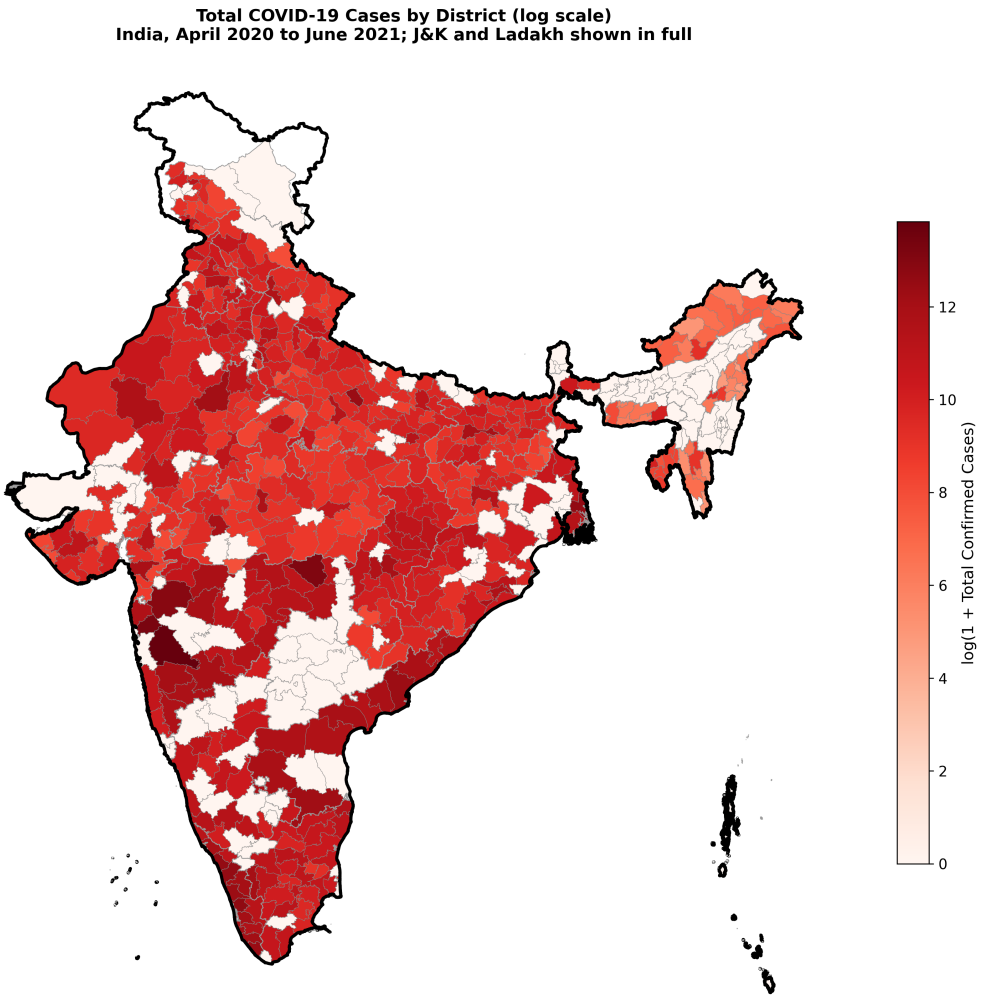


Figure 5: Total confirmed COVID-19 cases by district (log scale), April 2020 to June 2021. White districts have no case data in the source dataset (144 of 666 districts in total, including the island UTs of Andaman & Nicobar and Lakshadweep and 6 J&K districts: Badgam, Bandipore, Kargil, Leh, Poonch, Shopian). Maharashtra, Kerala, Karnataka, Tamil Nadu, and Uttar Pradesh show the heaviest burden. J&K districts with data (Srinagar, Baramulla, Jammu, etc.) are clearly visible in the north-west.

Figure 5 shows the geographic distribution of total COVID-19 cases. High-burden districts cluster in:

- Maharashtra (Mumbai, Pune, Thane): early importation via air travel
- Kerala: early detection and reporting capacity
- Karnataka and Tamil Nadu: major urban centres
- Uttar Pradesh districts: high population density amplification

J&K districts with available data are clearly visible (Srinagar, Jammu, Baramulla, Anantnag). Districts with null data (Kargil, Leh, Badgam, Bandipore, Poonch, Shopian) are shown as no-data (light grey) and retained as structural nodes in the network analysis without imputation.

5 Interactive Dashboard

An interactive HTML dashboard (`critical_nodes_simulator.html`) accompanies this report. It is fully standalone (all data embedded as JSON, D3.js via CDN) and works when opened locally or served via HTTP. The dashboard provides five scenario tabs:

1. **COVID Spread (SIR)**: An interactive SIR epidemic simulation seeded by clicking any district. The user can select the initial infected node; the simulation propagates day-by-day through the adjacency network. Colour indicates S/I/R state.
2. **Network Failure (Railway Resilience)**: Districts can be removed interactively to simulate node failures. The dashboard updates connected components in real time, showing how network fragmentation emerges as high-betweenness nodes are removed.
3. **Healthcare Access**: Colours districts by hop-distance to the nearest major medical hub, visualising healthcare access inequality across India's geographic network.
4. **Disaster Relief Logistics**: Simulates warehouse placement. Users designate supply depots; the dashboard shows hop-distances from each warehouse to all districts, identifying coverage gaps.
5. **Rural Connectivity**: Allows simulated edge addition to isolated rural districts, showing how a single new road connection can dramatically reduce effective isolation.

The map renders all 666 districts including J&K and Ladakh (Leh, Kargil). Centrality views (degree, betweenness, closeness, eigenvector) and Louvain community colouring are available via the sidebar controls on each scenario tab.

6 Discussion

6.1 Why the Siliguri Corridor Dominates

The Siliguri Corridor is a classic example of a *bridge node* in a near-planar network: removing its 4 districts disconnects 7 states (and ≈ 45 million people in North-East India)

from the remainder of the network. High betweenness centrality in geographic networks does not require being a large or densely populated node. Rather, it requires being a bottleneck in the graph's shortest-path structure. This has direct policy implications: epidemic early warning systems, supply chain resilience planning, and healthcare resource allocation should all treat the Siliguri Corridor as a structurally priority region.

6.2 Topology: Not Scale-Free, But Not Random

India's district network has a unimodal degree distribution with mean ≈ 5. This lattice-like topology arises because geographic adjacency networks are constrained by Euler's formula for planar graphs: no planar graph can have average degree exceeding 6. The network is also not random: average clustering (0.46) is far higher than expected in an Erdős–Rényi graph with the same density, reflecting the local planarity of geographic boundaries.

6.3 Limits of Adjacency as a Contact Proxy

The negative but modest correlations between centrality and timing ($|r| < 0.30$) reflect genuine omitted variables: air travel from Mumbai and Delhi (non-adjacent to most districts) drove early seeding in distant cities; pilgrimage events (Kumbh Mela, Nizamuddin) created super-spreader hubs not captured by border adjacency; population density within districts affects transmission after importation. A richer model would weight edges by inter-district mobility data.

A second source of attenuation is data left-censoring. The covid19india.org district-level series begins in April 2020, while the clock for time-to-first-case is set at India's index case on 30 January 2020. Districts infected in February or March 2020 — including major hubs such as Mumbai, Delhi, and several in Kerala — cannot show a time-to-first-case below approximately 86 days; their early arrival is censored up to the April reporting floor, visible as the hard lower bound in Figure 3. This compresses the early variance of the outcome and attenuates the centrality–timing correlations independently of the omitted-variable concerns above.

6.4 J&K and Ladakh: A Note on Administrative Reorganisation

The analysis treats Leh (Ladakh) and Kargil as districts of the pre-2019 Jammu & Kashmir, consistent with the GADM boundary data available. Post-2019, these districts belong to the Ladakh Union Territory. Their structural properties in the network remain unchanged by this reclassification: Kargil has degree 5 and betweenness 0.015 (moderate

connectivity within the Himalayan fringe), while Leh has degree 1 (near-isolated, as expected for a high-altitude district bordered primarily by Tibet and Kargil). Both lack COVID-19 case data in the source dataset and are flagged as NaN throughout.

7 Conclusion

India's 666-district geographic adjacency network is a lattice-like, near-planar graph whose topology directly shaped early COVID-19 propagation. The key findings are:

1. The network is **not scale-free**; degree distribution is unimodal (mean = 5.36), as expected for a geographically embedded lattice.
2. **Closeness centrality is the best predictor** of time-to-first-COVID-case ($r = -0.28$), followed by degree ($r = -0.20$). Being geometrically close to all parts of India predicts early infection better than being a transit hub.
3. **The Siliguri Corridor** (Darjeeling, Jalpaiguri, Koch Bihar, Uttar Dinajpur) is India's dominant structural chokepoint, with betweenness values 40 to 65% higher than any other region. It is the sole land bridge to North-East India.
4. **Louvain communities** (13 detected) follow geographic and terrain features, crossing state administrative lines, implying that epidemiological surveillance zones should be defined geographically, not administratively.
5. **J&K and Ladakh** are correctly represented in the network (22 districts); 6 districts have missing COVID data and are retained as structural nodes with null case counts.

These results operationalise a practical recommendation: *epidemic early-warning sentinel nodes should be placed at high-closeness districts and at structural chokepoints like the Siliguri Corridor*, not simply at high-population urban centres.

Acknowledgements

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References

[1] Moritz UG Kraemer, Chia-Hung Yang, Bernardo Gutierrez, Chieh-Hsi Wu, Brennan Klein, David M Pigott, Open COVID-19 Data Working Group, Louis Du Plessis, Nuno R Faria, Ruoran Li, et al. The effect of human mobility and control measures on the covid-19 epidemic in china. *Science*, 368(6490):493–497, 2020. doi: 10.1126/science.abb4218.

[2] Ulrik Brandes. A faster algorithm for betweenness centrality. In *Journal of Mathematical Sociology*, volume 25, pages 163–177, 2001. doi: 10.1080/0022250X.2001.9990249.

[3] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008. doi: 10.1088/1742-5468/2008/10/P10008.

[4] Aric A Hagberg, Daniel A Schult, and Pieter J Swart. Exploring network structure, dynamics, and function using NetworkX. pages 11–15, 2008.